

Problem 4

a) In a traditional knockout tournament with n rounds, there are 2^n players and $M = 2^n - 1$ matches as in every match one player is eliminated until one player (the winner of the tournament) is left without an opponent.

Let $m \in \{1, \dots, M\}$ an index to identify each match. A sequence

$\omega = (\omega_1, \dots, \omega_M)$ represents the winner ω_m of the match m .

Then, $\Omega = \{\omega \mid \omega \text{ is a valid progression in the tournament}\}$. A progression is valid iff a winner in the round k was the corresponding winner in the round $k-1$. The σ -algebra $\mathcal{F} = \mathcal{P}(\Omega)$.

Since the winning chances of two players i and j in a match stand at $s_i : s_j$, then the probability of player i of winning a match against player j is $\frac{s_i}{s_i + s_j}$. Then the probability of a specific progression ω

to become the outcome of the tournament is the product of all probabilities of the individual results (assuming independence of matches given the participants): $\mathbb{P}(\{\omega\}) = \prod_{m=1}^M \frac{s_{\omega_m}}{s_{i(m)} + s_{j(m)}}$ (✓) where s_{ω_m} is

the strength of the winner of match m , $s_{i(m)}$ and $s_{j(m)}$ are the strength of the players participating in match m of path ω .

b) The model can be simplified. If all players have the same strength, every player has the same probability ($\frac{1}{2^n}$) of winning. If we are only interested in the winner, then the sample space is simplified. Specifically:

$$\Omega' = \{1, \dots, 2^n\} \quad (\text{the players}),$$

$$\mathcal{F}' = \mathcal{P}(\Omega'),$$

$$\mathbb{P}(\{i\}) = \frac{1}{2^n} \quad \forall i \in \Omega' \quad \checkmark$$

c) is not sufficient to exclude tuples such as $(1, 1, \dots, 1)$, which is invalid.

Furthermore, in the definition of $\mathbb{P}$ the loser of the m th match is required.

How can one access this information from ω (it is possible, but not trivial)

a) $\frac{1}{5}$

b) $\frac{1}{4}$

$\frac{2}{6}$